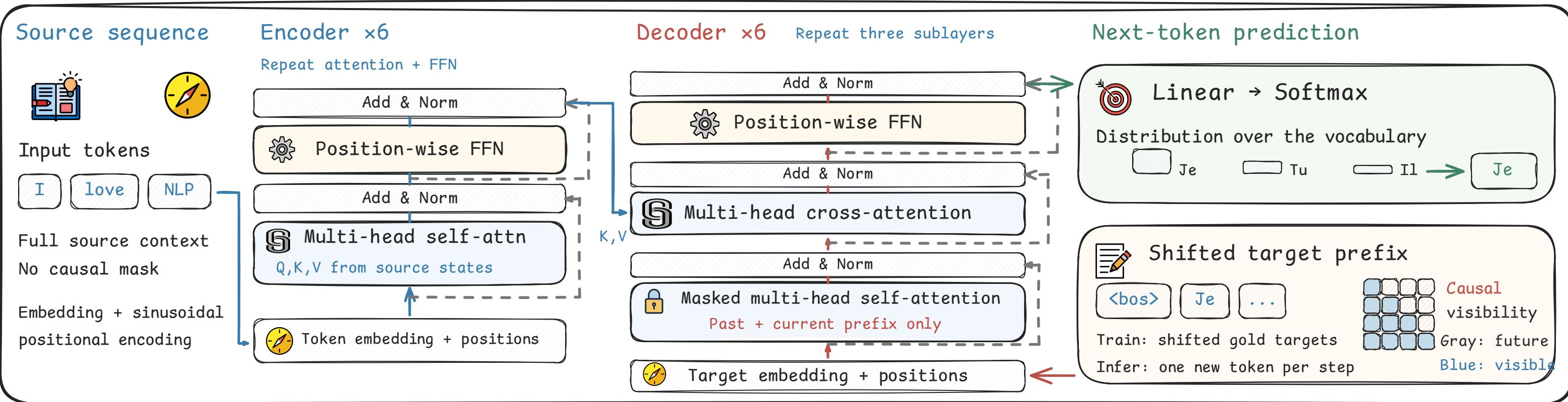


Transformer

Attend globally · preserve order · decode causally



Scaled dot-product attention

Q: queries K: keys V: values

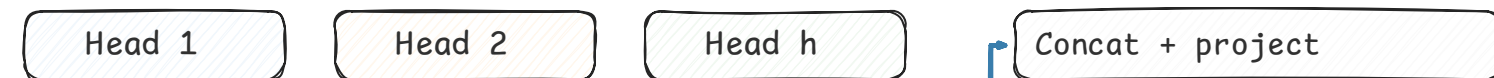
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right) V$$



Cross: Q from decoder; K, V from encoder. Padding is masked.



Multiple heads + feed-forward transformation



$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$



$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$



$$\text{Add \& Norm}(x) = \text{LayerNorm}(x + \text{Sublayer}(x))$$

8 heads · width 512 · FFN 2048 · separate Q/K/V projections